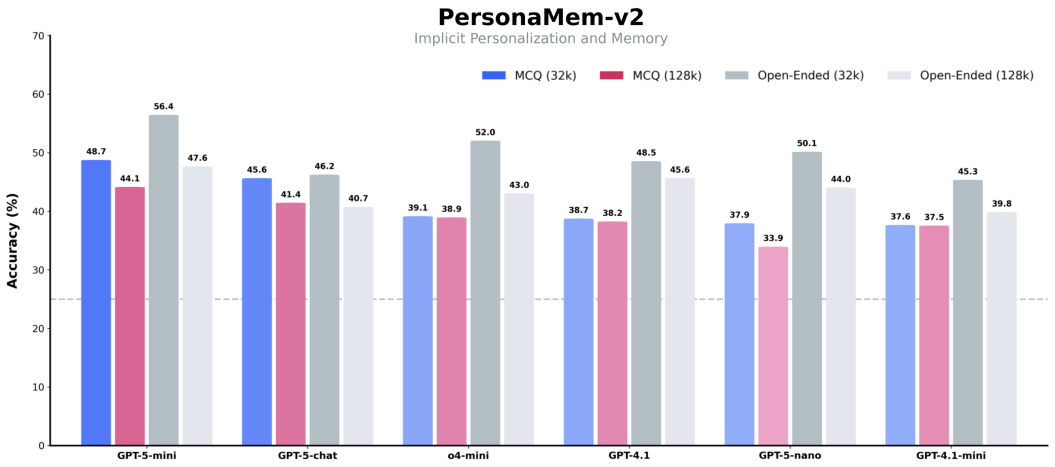

Figure 4: Performance of OpenAI models on the PERSONAMEM-V2 benchmark, comparing accuracy across 32k and 128k input contexts in MCQ and Open-Ended tasks. Despite recent advancements, we observe that frontier models still struggle with implicit personalization. The dashed line represents the random guess performance in MCQ tasks.

These constraints force the model to decide what to store without knowing future user queries, encouraging it to anticipate what information will matter and to extract the essential user persona. At each step, the model reads the current chunk C_i and the previous memory M_{i-1} to produce an updated memory. After processing all chunks, the same model answers the final query q using the refined memory M_T. The pipeline is shown in Figure 3. Formally:

$$M_i = f_\theta(C_i,\, M_{i-1}), \quad \hat{y} = f_\theta(M_T,\, q)$$

where f_θ is the language model, C_i is the i-th chunk out of T, M_{i-1} and M_i are the previous and updated memories, q is the current user query, and $\hat{y}$ is the model response to q.

We use a single LLM for the entire process: the same model that writes the memory also gives the final answer. All queries in PERSONAMEM-V2 can only be answered in a personalized manner by having a correct user understanding, so the model will receive high rewards if the memory turns out to be both necessary and helpful in giving correct personalized responses, and vice versa. Over time, **the model learns to reason over context and maintain a concise, human-readable memory that grows with each user over time**.

4 Experimental Results

4.1 Benchmarking Frontier LLMs in Personalization

We benchmark a series of OpenAI models on our personalization suite, including GPT-5-Chat, GPT-5-mini, GPT-5-nano, GPT-4.1, GPT-4.1-mini, and o4-mini (OpenAI, 2025a; 2024b). Each model is evaluated under both multiple-choice (MCQ) and open-ended settings. For open-ended evaluations, we adopt an LLM-as-a-judge protocol: three independent GPT-5-Chat instances score each response, and we aggregate their judgments to obtain more robust accuracy estimates. The overall results are shown in Figure 4.

4.1.1 Frontier LLMs still struggle to infer implicit user preferences

Despite major advances in long-context handling, frontier LLMs still perform poorly when required to infer implicit user preferences. Across both MCQ and open-ended settings, which show strong correlation, GPT-5 variants reach only 40-55% accuracy. This reveals an obvious gap in current frontier models' ability to understand user preferences and offer personalized responses based on subtle cues in interaction history.

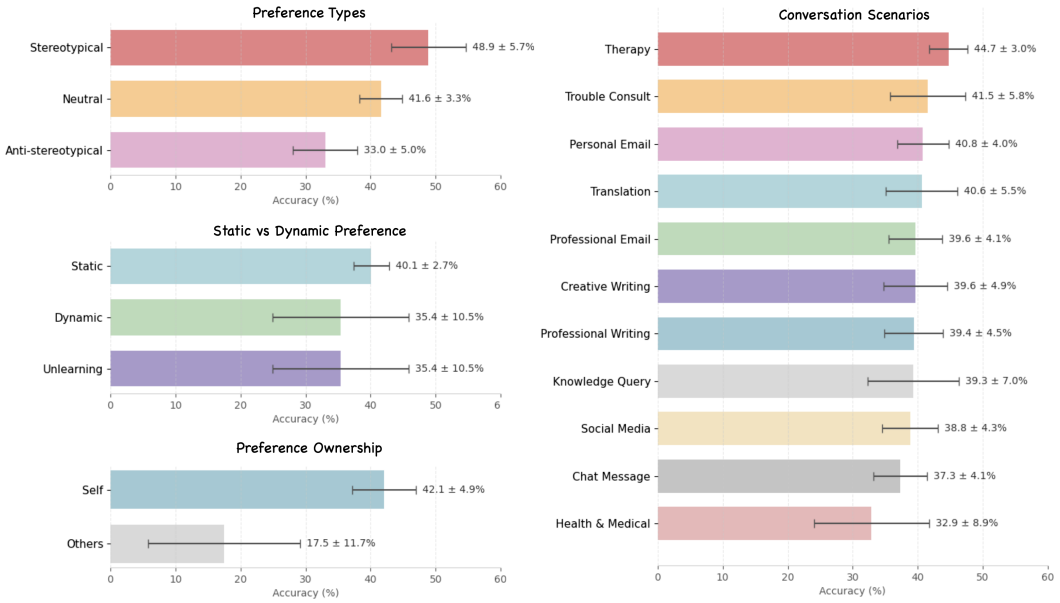

Figure 5: Breakdown of model accuracy by preference types and scenarios, aggregated across all evaluated models. The results indicate systematic variance in personalization capabilities, showing a reliance on population priors, with performance dropping for anti-stereotypical and dynamic preferences and distinguishing preference ownerships.

4.1.2 Reasoning, not long-context capabilities, drives success in implicit personalization

The models that lead the benchmark like GPT-5-mini and o4-mini are those with strong reasoning capabilities. Meanwhile, perhaps surprisingly, we observe no significant accuracy improvement when the context is shortened from 128k to 32k tokens by removing conversations irrelevant to the current persona. These results suggest that the main bottleneck in implicit personalization is not simply memorizing longer histories or retrieving factual information, but correctly interpreting and integrating subtle preference signals embedded within those histories. We need to update our focus: moving from "needle-in-a-haystack" (Kamradt, 2023; Team et al., 2024) retrieval-style stress tests toward more nuanced, fine-grained assessments that reflect the demands of real-world personalization.

4.1.3 Implicit personalization varies systematically across preference types

Figure 5 shows that implicit personalization varies systematically across preference types. Frontier models perform best when user preferences align with common stereotypes, achieving 48.9 ± 5.7% accuracy aggregated across all models and context window lengths, but it drops to 41.6 ± 3.3% for neutral preferences and further to 33.0 ± 5.0% for anti-stereotypical preferences, suggesting reliance on population priors rather than individual behavior.

Besides, static preferences reach 40.1 ± 2.7% accuracy, while dynamic preferences fall to 35.4 ± 10.5%, indicating instability and difficulty updating beliefs as preferences change. For preference ownership, models infer users' own preferences correctly at 42.1 ± 4.9%, but perform far worse, only 17.5 ± 11.7%, when distinguishing preferences of others from the user's own. Differences across task scenarios are comparatively modest, ranging from 44.7 ± 3.0% in therapy consultation to 32.9 ± 8.9% in physical health and medical scenarios.

4.2 Training Long-context Reasoning for Personalization

We start from Qwen3-4B-Instruct-2507 model (Yang et al., 2025), trained with verl: Volcano Engine Reinforcement Learning for LLMs (Sheng et al., 2025) and vLLM (Kwon et al., 2023) on 8 Nvidia H100 GPUs. The training dataset contains 18,000 samples with no persona

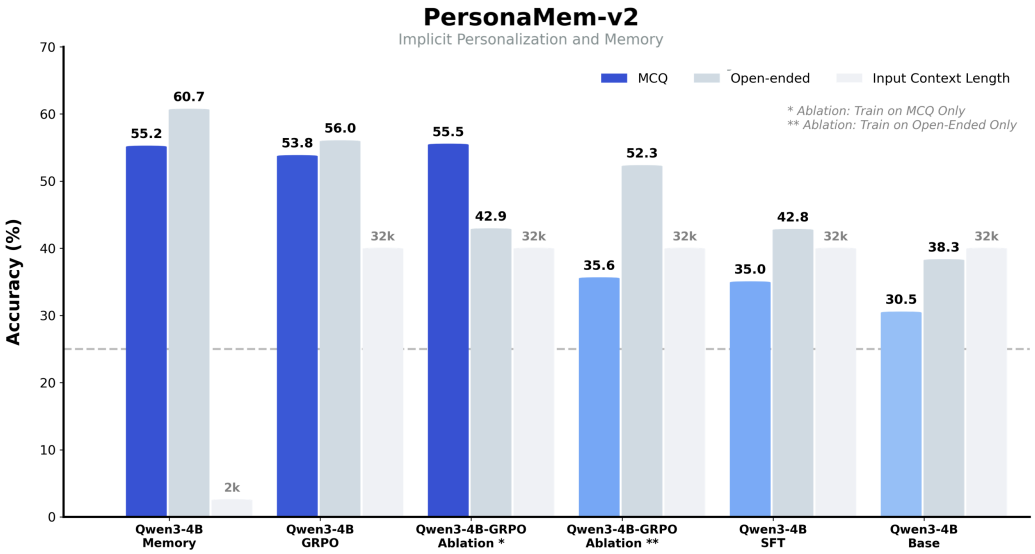

Figure 6: Performance of our Qwen3-4B models on PERSONAMEM-V2 trained via SFT, GRPO, and Agentic Memory, alongside ablation studies for GRPO trained solely on MCQ or Open-Ended data. Comparisons show that RL drives significant gains in implicit personalization. Notably, our Agentic Memory framework achieves both SOTA performance and efficiency using only 2k input tokens as memory throughout the 32k token history. The dashed line represents the random guess performance in MCQ tasks.

overlap with the benchmark set. All models are trained on this dataset and evaluated on the benchmark set, using a context window of up to 32k tokens. The model is first cold-started with 300 steps of supervised fine-tuning with a batch size of 64 on our training data, then reinforced with GRPO for 1 epoch around 500 steps, enabling thinking and using a batch size of 32 with 8 rollouts. We mix 80% multiple-choice and 20% open-ended queries to keep training stable with more verifiable rewards from MCQ while still preserving open-ended conversational abilities for real-world personalization. We use three GPT-5-Chat instances as LLM-as-judges to provide rewards on the open-ended samples.

4.2.1 Reinforcement learning incentives reasoning toward personalization

While personalization is often viewed as subjective and pluralistic, we find that it can still be effectively incentivized through reinforcement fine-tuning. In particular, well-designed reward signals encourage models to reason about whether a response aligns with or violates the current user's persona and preferences. Our dataset plays a crucial role here: it provides the structured yet diverse supervision needed for RL to learn these behaviors. Using this data, our 4B-parameter reasoning model, marked as Qwen3-4B-GRPO in Figure 6, gains substantial improvements after post-training, reaching 53.8% on MCQ and 56.0% on open-ended tasks, superior to supervised fine-tuning. Notably, this small model even outperforms GPT-5-Chat, which achieves 45.6% and 46.2% on the same benchmarks. These results show that reinforcement learning can meaningfully shape a model's ability to interpret and integrate subtle preference signals to give personalized responses.

4.2.2 Hybrid reward signals unlock better RL toward personalization

To understand how different supervision signals shape reinforcement fine-tuning for personalization, we ran ablations comparing the same Qwen3-4B models trained only on MCQ, only on open-ended questions, or on our mixed dataset, keeping the same training setup, GRPO algorithm, and total number of training samples. The results shown in Figure 6 are striking: the MCQ-only model collapsed by 13.1% on open-ended personalization tasks needed for real-world user-facing services, while the open-ended–only model dropped